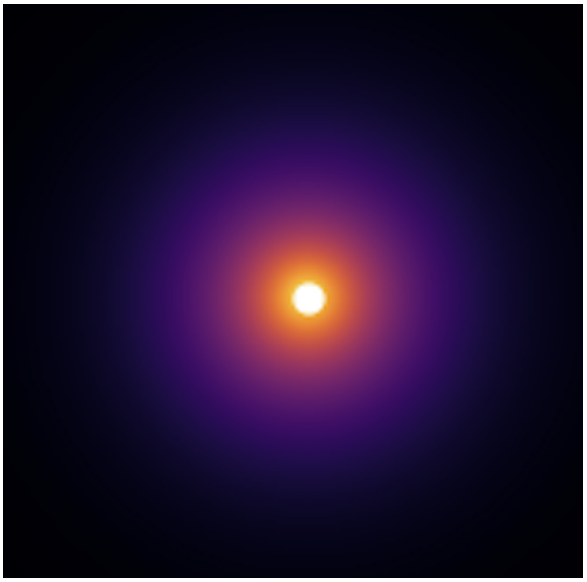
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The Art of Scientific Computing: A Matter of Gaussian Approximation

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Contents

1 Introduction **2**
- 1.1 Hydrogenic Atoms . . . 2
- 1.2 Gaussian Basis Sets . . . 3
- 1.3 Accuracy and Limitations of Gaussian Expansions . . . 4
- 1.4 Motivation and Scope . . . 4

2 Theory **6**
- 2.1 Exact Hydrogenic Ground State . . . 6
- 2.2 Gaussian Basis and Variational Method . . . 7
- 2.3 Accuracy, Conditioning and Nuclear-Charge Scaling . . . 8

3 Computational Methodology **10**
- 3.1 Variational Calculation . . . 10
- 3.2 Investigating Accuracy and Basis Limitations . . . 10
- 3.3 Numerical Verification and Stability . . . 12

4 Results and Discussion **13**
- 4.1 Hydrogen Benchmark . . . 13
- 4.2 Basis-Size Convergence and Exponent Selection . . . 15
- 4.3 Wavefunction Accuracy and Near-Nuclear Behaviour . . . 17
- 4.4 Transferability and Nuclear-Charge Scaling . . . 20

5 Conclusion and Limitations **23**
- 5.1 Main Findings . . . 23
- 5.2 Limitations and Further Work . . . 24

A Derivation of Gaussian Matrix Elements **26**
- A.1 Gaussian Normalisation . . . 26
- A.2 Overlap Matrix . . . 26
- A.3 Kinetic-Energy Matrix . . . 27
- A.4 Electron–Nucleus Potential Matrix . . . 27
- A.5 Variational Eigenvalue Problem . . . 28

Bibliography **28**

Chapter 1

Introduction

Accurately describing the behaviour of electrons is fundamental to understanding atomic structure, chemical bonding and the properties of materials. However, most atomic and molecular systems cannot be solved analytically, making numerical approximations essential in computational physics and chemistry. Hydrogen and hydrogenic ions provide an ideal starting point for investigating the reliability of these methods, as their exact non-relativistic energies and wavefunctions are known. This allows the accuracy of an approximation to be measured directly rather than inferred from numerical convergence alone. Importantly, reproducing an accurate energy does not necessarily guarantee an equally accurate wavefunction, while increasing the complexity of an approximation can introduce additional numerical difficulties. These considerations highlight the importance of constructing approximations that balance physical accuracy, numerical reliability and computational efficiency. Such methods underpin modern computational chemistry, with applications ranging from pharmaceutical research, where quantum-mechanical calculations help investigate drug–receptor interactions and chemical reactivity [8], to astrochemistry, where predicted molecular properties and spectra help identify complex and potentially prebiotic molecules in interstellar environments [3].

1.1 Hydrogenic Atoms

Hydrogenic atoms contain a single electron bound to a nucleus of charge $+Ze$, where Z is the atomic number. Hydrogen ($Z = 1$) is the simplest example, but the same model also describes one-electron ions such as He^{+}, Li^{2+} and Mn^{24+}. Unlike multi-electron atoms, hydrogenic systems have no electron–electron interactions. Their non-relativistic Schrödinger equation can therefore be solved analytically, providing exact wavefunctions and energies against which numerical approximations can be tested.

In atomic units, assuming an infinitely massive point nucleus, the Hamiltonian is

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{Z}{r}, \tag{1.1}$$

where r is the electron–nucleus separation. The exact ground-state energy and normalised $1s$ wavefunction are

$$E_{1s} = -\frac{Z^2}{2}, \qquad \psi_{1s}(r) = \sqrt{\frac{Z^3}{\pi}} e^{-Zr}. \tag{1.2}$$

As Z increases, the electron becomes more tightly bound and its wavefunction becomes more concentrated near the nucleus. The ground-state energy scales as Z^2, while the characteristic radial length scale decreases as $1/Z$.

Hydrogenic systems are therefore useful for examining the accuracy, convergence and numerical stability of approximate quantum-mechanical methods. Since the exact solution is known, errors can be measured directly without uncertainty in the reference values.

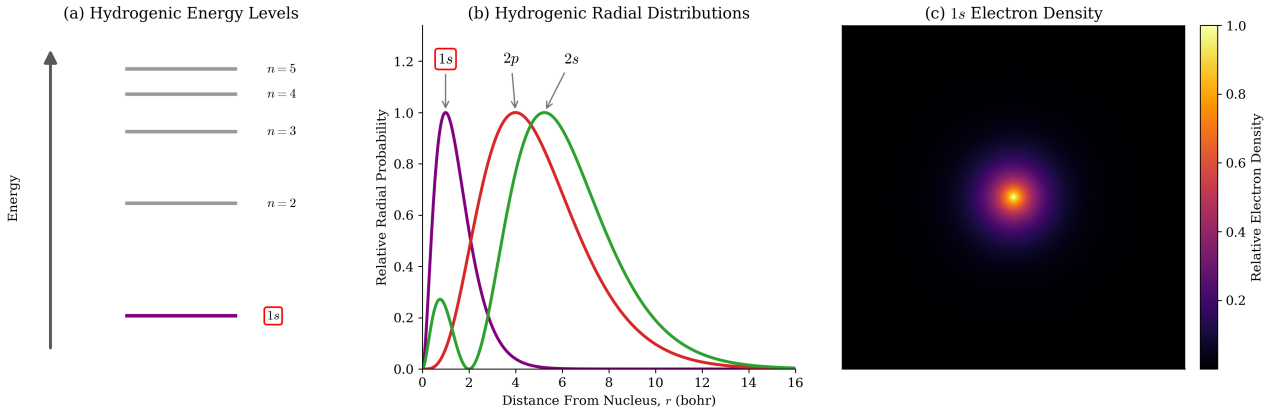


Figure 1.1: Overview of the hydrogenic atom: (a) discrete electronic energy levels, (b) radial probability distributions for the $1s$, $2s$ and $2p$ orbitals, and (c) the spatial electron density of the $1s$ orbital. The focus here is on the ground state, for which exact results provide a reference for numerical calculations.

1.2 Gaussian Basis Sets

Although the hydrogenic Schrödinger equation has an exact solution, most atomic and molecular systems cannot be solved analytically. Numerical electronic-structure methods therefore approximate wavefunctions using finite combinations of simpler functions, known as basis functions.

One common choice is the Gaussian function, introduced into electronic-structure calculations by Boys [4]. A spherical Gaussian has the form

$$\phi_i(r) = A_i e^{-\lambda_i r^2}, \tag{1.3}$$

where $\lambda_i > 0$ determines its spatial width and A_i is a normalisation constant. Small exponents give broad functions extending farther from the nucleus, while large exponents give narrow functions concentrated near it.

The hydrogenic ground-state wavefunction can be approximated by a linear combination of N Gaussians,

$$\psi_{\mathrm{G}}(r) = \sum_{i=1}^{N} c_i \phi_i(r), \tag{1.4}$$

where the coefficients c_i determine the contribution of each function. These coefficients are found by minimising the energy expectation value, leading to a generalised eigenvalue problem. Its eigenvalues give approximate energies, while its eigenvectors determine the corresponding wavefunctions.

The main computational advantage of Gaussian functions is that their products and many of the required integrals can be evaluated analytically. This makes them useful for constructing the Hamiltonian and overlap matrices in electronic-structure calculations [4]. Gaussian basis sets are widely used in quantum chemistry, including in contracted and correlation-consistent basis sets [6, 13].

The choice of Gaussian exponents determines the range of distances that the basis can represent. Here, the exponents follow a geometric sequence,

$$\lambda_i = \alpha\beta^{i-1}, \qquad i = 1, \ldots, N, \tag{1.5}$$

where $\alpha > 0$ sets the smallest exponent, $\beta > 1$ controls the spacing between successive exponents, and N is the number of functions.

This is known as an *even-tempered Gaussian basis*. Each exponent is β times the previous one, giving equal spacing on a logarithmic scale. Changing N, α and β therefore allows the number and distribution of functions to be adjusted systematically. Even-tempered bases and Gaussian exponent optimisation have been studied by Feller and Ruedenberg [7], Bardo and Ruedenberg [2], and Wilson [14].

1.3 Accuracy and Limitations of Gaussian Expansions

Despite their computational advantages, Gaussian functions do not have the same radial behaviour as exact hydrogenic wavefunctions. The exact $1s$ wavefunction decreases as e^{-Zr}, whereas an individual Gaussian decreases as $e^{-\lambda r^2}$. A finite sum of Gaussians can approximate the exact wavefunction closely over a range of distances, but cannot reproduce it exactly everywhere.

An important difference occurs at the nucleus. The exact hydrogenic wavefunction satisfies the electron–nucleus cusp condition:

$$\left.\frac{\mathrm{d}\psi_{1s}}{\mathrm{d}r}\right|_{r=0} = -Z\psi_{1s}(0), \tag{1.6}$$

whereas the radial derivative of any finite sum of centred spherical Gaussians is zero at $r = 0$. This limitation is related to the Coulomb interaction near the nucleus [9, 10].

This leads to an important distinction between energy and wavefunction accuracy. The variational method gives an upper bound to the exact ground-state energy, but a small energy error does not mean that every feature of the wavefunction is equally accurate. In particular, the behaviour near the nucleus may still differ noticeably from the exact solution.

Adding more Gaussians generally makes the basis more flexible, although the improvement depends on the chosen exponents. Functions with very similar widths can become nearly linearly dependent, making the overlap matrix poorly conditioned and the numerical solution less reliable [11]. A useful Gaussian basis therefore cannot be judged by its size alone. Energy accuracy, wavefunction accuracy and numerical conditioning must all be considered.

1.4 Motivation and Scope

Gaussian basis sets are widely used in computational chemistry, particularly in electronic-structure methods such as Hartree–Fock theory and density functional theory [1]. These methods approximate the electronic states of atoms and molecules, allowing properties such as molecular energies, charge distributions and chemical bonding to be investigated. Such calculations contribute to materials modelling, the interpretation of electronic spectra and pharmaceutical research, where understanding molecular stability, reactivity and intermolecular interactions is important. However, the accuracy of these predictions depends partly on the number and choice of basis functions. Too few functions, or poorly chosen exponents, may give inaccurate energies and electron distributions, while adding more functions increases computational cost and can introduce numerical instability. Choosing a suitable basis therefore involves balancing accuracy, flexibility and numerical reliability.

The hydrogenic ground state provides a useful way to examine this problem because its exact energy and wavefunction are known. Errors in the Gaussian approximation can therefore be measured directly, rather than relying only on whether a calculation appears to have converged. This raises the central question:

How many Gaussian functions are needed to reproduce a hydrogenic ground state to a chosen accuracy while maintaining numerical reliability?

The first step is to compare the approximate hydrogen energy, wavefunction and radial probability distribution with their exact values. Accuracy is then examined as the number of Gaussians and the exponent parameters α and β are varied. The overlap condition number provides a measure of numerical reliability, while a parameter search identifies the smallest tested bases meeting selected energy-error targets. Comparing energy and wavefunction errors also shows whether an accurate energy is accompanied by an equally accurate electron distribution.

A further question is whether a basis chosen for hydrogen remains suitable for one-electron ions with larger nuclear charges. As Z increases, the electron distribution becomes more concentrated near the nucleus, so the original hydrogen exponents may no longer cover the required range of distances. To test this, fixed hydrogen exponents are compared with those scaled by

$$\lambda_i(Z) = Z^2\lambda_i(1), \tag{1.7}$$

which adjusts the Gaussian widths to match the hydrogenic length scale. The comparison covers $1 \leq Z \leq 25$, including Mn^{24+}, and considers both energy accuracy and radial probability distributions.

The aim is not simply to obtain an accurate ground-state energy, but to understand why some Gaussian bases perform better than others, how numerical reliability limits their usefulness, and how the basis must be adapted to the physical length scale of the system. These questions are relevant to basis-set selection in computational chemistry, although the conclusions from a one-electron model do not directly account for the additional complexity of many-electron systems.

All calculations use the non-relativistic, fixed-nucleus, one-electron Hamiltonian with spherical Gaussians centred on the nucleus. Only ground states are considered. Electron–electron interactions, molecular geometry and relativistic effects are outside the scope of this investigation.

Chapter 2 develops the theoretical framework, Chapter 3 explains the numerical methods, and Chapter 4 presents and discusses the results. Chapter 5 summarises the main findings and limitations.

Chapter 2

Theory

The numerical investigation undertaken requires a theoretical framework against which the Gaussian approximation can be assessed. This chapter begins with the exact hydrogenic ground state, which provides a benchmark for evaluating the approximate solutions. The variational formulation and analytical Gaussian matrix elements are then developed, followed by the measures of accuracy, numerical conditioning and nuclear-charge scaling used throughout the subsequent analysis.

2.1 Exact Hydrogenic Ground State

In atomic units ($\hbar = m_e = e = 4\pi\varepsilon_0 = 1$), the Hamiltonian for an electron bound to a fixed point nucleus of charge $+Ze$ is

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{Z}{r}, \tag{2.1}$$

where the two terms represent the electron's kinetic energy and its attractive Coulomb interaction with the nucleus. The exact normalised ground-state wavefunction and corresponding energy are [5]

$$\psi_{1s}(r) = \left(\frac{Z^3}{\pi}\right)^{1/2} e^{-Zr}, \qquad E_{1s} = -\frac{Z^2}{2}. \tag{2.2}$$

The probability density $|\psi(r)|^2$ gives the probability per unit volume of finding the electron at a particular position. For a spherically symmetric state, the radial probability density is

$$P(r) = 4\pi r^2 |\psi(r)|^2, \qquad \int_0^\infty P(r)\,\mathrm{d}r = 1. \tag{2.3}$$

The factor $4\pi r^2$ accounts for the volume of a spherical shell. Substituting the exact $1s$ wavefunction gives

$$P_{1s}(r) = 4Z^3 r^2 e^{-2Zr}.$$

Although the wavefunction is largest at the nucleus, the radial probability reaches its maximum at $r = 1/Z$. The mean radius is $\langle r \rangle = 3/(2Z)$, reflecting the contribution from the full radial distribution rather than only its most probable radius.

These exact results provide complementary benchmarks for the Gaussian approximation. The ground-state energy measures the quality of the variational result, while the wavefunction, radial probability and mean radius describe its spatial accuracy. Both are needed because an accurate energy does not necessarily imply an equally accurate wavefunction.

2.2 Gaussian Basis and Variational Method

The approximate ground-state wavefunction is represented by a linear combination of N spherical Gaussian functions centred on the nucleus [12],

$$\psi_{\mathrm{G}}(r) = \sum_{i=1}^{N} c_i \phi_i(r), \qquad \phi_i(r) = A_i e^{-\lambda_i r^2}, \qquad A_i = \left(\frac{2\lambda_i}{\pi}\right)^{3/4}. \tag{2.4}$$

The coefficients c_i determine the contribution of each function, while A_i normalises each Gaussian individually. The exponents follow an even-tempered sequence,

$$\lambda_i = \alpha\beta^{i-1}, \qquad \alpha > 0, \quad \beta > 1, \quad i = 1, \ldots, N. \tag{2.5}$$

Here α sets the smallest exponent and β determines the ratio between successive exponents [12, 2, 7, 14].

The characteristic width of each Gaussian is proportional to $1/\sqrt{\lambda_i}$. Small exponents describe more diffuse regions of the electron distribution, while large exponents produce narrower functions concentrated near the nucleus. Since the largest exponent is $\alpha\beta^{N-1}$, increasing N at fixed α and β adds progressively narrower functions without extending the diffuse end of the basis. Consequently, increasing the number of functions does not necessarily improve the description of every radial region equally.

Gaussian functions are particularly useful because their products and many of the required integrals can be evaluated analytically. The Gaussian product theorem expresses the product of two Gaussians as another Gaussian, simplifying the construction of the matrices used in electronic-structure calculations [12]. Although the exact hydrogenic wavefunction decays as e^{-Zr} rather than $e^{-\lambda r^2}$, a suitable combination of Gaussians can approximate its spatial behaviour while retaining these computational advantages.

The expansion coefficients are obtained by minimising the expectation value of the Hamiltonian. For a nonorthogonal basis, the energy is expressed through the Rayleigh quotient,

$$E[\mathbf{c}] = \frac{\mathbf{c}^{\mathsf{T}}\mathbf{H}\mathbf{c}}{\mathbf{c}^{\mathsf{T}}\mathbf{S}\mathbf{c}}, \qquad H_{ij} = \langle\phi_i|\hat{H}|\phi_j\rangle, \quad S_{ij} = \langle\phi_i|\phi_j\rangle. \tag{2.6}$$

The Hamiltonian matrix $\mathbf{H}$ contains the kinetic and potential energy contributions, while the overlap matrix $\mathbf{S}$ accounts for the nonorthogonality of the Gaussian basis functions.

According to the variational principle, the expectation value of the Hamiltonian for any normalised trial wavefunction cannot be lower than the exact ground-state energy [5]. The coefficients are therefore chosen to minimise the energy subject to $\mathbf{c}^{\mathsf{T}}\mathbf{S}\mathbf{c} = 1$. Introducing the Lagrange multiplier E, identified with the stationary energy, gives

$$\mathcal{L} = \mathbf{c}^{\mathsf{T}}\mathbf{H}\mathbf{c} - E\left(\mathbf{c}^{\mathsf{T}}\mathbf{S}\mathbf{c} - 1\right).$$

Differentiating with respect to the coefficient vector,

$$\frac{\partial\mathcal{L}}{\partial\mathbf{c}} = 2\mathbf{H}\mathbf{c} - 2E\mathbf{S}\mathbf{c} = 0,$$

leads to the generalised eigenvalue equation

$$\mathbf{H}\mathbf{c} = E\mathbf{S}\mathbf{c}. \tag{2.7}$$

This formulation is also used in the computational quantum-chemistry approach described by Quiney and Flynn [12]. Further details are given in Appendix A.

The overlap matrix appears because the Gaussian functions are not mutually orthogonal, so $\mathbf{S} \neq \mathbf{I}$. For a linearly independent basis, $\mathbf{S}$ is positive definite and the generalised eigenvalues are real. The lowest eigenvalue gives the minimum energy available within the chosen Gaussian space. The variational bound therefore requires $E_{\mathrm{G}} \geq E_{1s}$ in exact arithmetic, providing a useful check on the numerical solution.

The matrix elements can be evaluated analytically. Defining $p_{ij} = \lambda_i + \lambda_j$, the overlap, kinetic-energy and

Coulomb potential matrices are

$$S_{ij} = A_i A_j \left(\frac{\pi}{p_{ij}}\right)^{3/2}, \tag{2.8}$$

$$T_{ij} = \frac{3\lambda_i \lambda_j}{p_{ij}} S_{ij}, \tag{2.9}$$

$$V_{ij} = -\frac{2\pi Z A_i A_j}{p_{ij}}, \tag{2.10}$$

$$H_{ij} = T_{ij} + V_{ij}. \tag{2.11}$$

These expressions agree with the one-centre Gaussian matrix elements described by Quiney and Flynn [12]. Their derivations, including Gaussian normalisation, are provided in Appendix A.

Evaluating the matrix elements analytically avoids numerical integration errors in the construction of $\mathbf{H}$ and $\mathbf{S}$. It also makes repeated calculations with different basis sizes and exponent parameters more efficient. The resulting generalised eigenvalue problem is solved numerically using the methods described in Chapter 3.

2.3 Accuracy, Conditioning and Nuclear-Charge Scaling

The primary measure of accuracy is the relative ground-state energy error,

$$\varepsilon_E = \frac{|E_{\mathrm{G}} - E_{1s}|}{|E_{1s}|}. \tag{2.12}$$

However, the variational energy alone does not establish how closely the approximate wavefunction reproduces the exact state. Spatial accuracy is therefore also assessed through the relative error in the mean radius,

$$\varepsilon_r = \frac{|\langle r\rangle_{\mathrm{G}} - 3/(2Z)|}{3/(2Z)}, \qquad \langle r\rangle_{\mathrm{G}} = \int_0^\infty r P_{\mathrm{G}}(r)\,\mathrm{d}r, \tag{2.13}$$

and the three-dimensional wavefunction error, expressed as an L^2 norm,

$$\varepsilon_\psi = \left[4\pi \int_0^\infty r^2 |\psi_{\mathrm{G}}(r) - \psi_{1s}(r)|^2\,\mathrm{d}r\right]^{1/2}. \tag{2.14}$$

Before calculating this difference, the overall sign of the numerical wavefunction is chosen to agree with the positive exact ground state.

The variational principle explains why the energy may be substantially more accurate than the wavefunction. For a normalised Gaussian approximation, write

$$\psi_{\mathrm{G}} = a\psi_{1s} + \delta\psi, \qquad \langle\psi_{1s}|\delta\psi\rangle = 0.$$

The exact ground state satisfies $\hat{H}\psi_{1s} = E_{1s}\psi_{1s}$. Using this relation and the normalisation of both states, the energy difference becomes

$$E_{\mathrm{G}} - E_{1s} = \langle\delta\psi|(\hat{H} - E_{1s})|\delta\psi\rangle. \tag{2.15}$$

The terms linear in the wavefunction difference cancel, leaving an energy-weighted quadratic expression. A small energy error therefore does not necessarily imply a comparably small wavefunction error. This distinction motivates the separate accuracy measures examined in Chapter 4.

Another limitation comes from the behaviour of the wavefunction at the nucleus. The exact hydrogenic wavefunction satisfies the electron–nucleus cusp condition, whereas a finite sum of centred spherical Gaussians has zero radial derivative at the origin:

$$\psi'_{1s}(0) = -Z\psi_{1s}(0), \qquad \psi'_{\mathrm{G}}(0) = 0. \tag{2.16}$$

The cusp follows from the singular Coulomb potential. For a spherically symmetric wavefunction,

$$\nabla^2\psi = \psi''(r) + \frac{2}{r}\psi'(r).$$

If $\psi'(0) \neq 0$, the second term contains a contribution proportional to $1/r$. For the exact hydrogenic state, the cusp condition ensures that this singular kinetic-energy contribution cancels the Coulomb potential as $r \to 0$. A finite Gaussian expansion cannot reproduce this cancellation exactly [9, 10].

The effect can be examined through the local energy,

$$E_{\mathrm{loc}}(r) = \frac{\hat{H}\psi(r)}{\psi(r)}. \tag{2.17}$$

For an exact eigenfunction, the local energy equals the corresponding eigenvalue wherever the wavefunction is nonzero. For a finite Gaussian expansion with nonzero amplitude at the nucleus, the kinetic-energy contribution remains finite as $r \to 0$, while the Coulomb term diverges as $-Z/r$. The local energy consequently tends towards negative infinity near the nucleus, revealing a limitation that may be less apparent from the total variational energy.

Increasing the number of Gaussians can improve the approximation, but it can also introduce numerical difficulties when basis functions become nearly linearly dependent. This is assessed using the spectral condition number of the overlap matrix,

$$\kappa(\mathbf{S}) = \frac{s_{\max}}{s_{\min}}, \tag{2.18}$$

where $s_{\max}$ and $s_{\min}$ are its largest and smallest eigenvalues. For a positive-definite overlap matrix, both are positive. A large condition number indicates that some combinations of basis functions are nearly redundant, making the generalised eigenvalue problem more sensitive to numerical rounding errors [11].

It is therefore important to distinguish between an incomplete basis and an ill-conditioned one. An incomplete basis may fail to represent the physical wavefunction accurately even when the matrix problem is solved correctly. Conversely, a larger basis may contain nearly dependent functions that make the numerical solution unreliable. In the parameter search, $\kappa(\mathbf{S}) \leq 10^{12}$ is used as a practical screening criterion rather than a universal guarantee of accuracy.

Hydrogenic wavefunctions also have a simple dependence on nuclear charge. The exact ground state depends on the scaled coordinate $\rho = Zr$, so its characteristic radial length decreases as $1/Z$ [5]. To preserve the Gaussian widths relative to this length scale, the exponents must satisfy

$$\lambda_i(Z) = Z^2 \lambda_i(1). \tag{2.19}$$

For an even-tempered basis, this corresponds to replacing α with $Z^2\alpha$ while keeping N and β unchanged.

The exponent scaling follows by writing each Gaussian in terms of ρ:

$$e^{-\lambda_i(Z) r^2} = e^{-[\lambda_i(Z)/Z^2]\rho^2}.$$

Choosing $\lambda_i(Z) = Z^2\lambda_i(1)$ therefore preserves the Gaussian shape in the scaled coordinate.

The corresponding energy scaling can be obtained directly from the Hamiltonian. Since $\nabla_r^2 = Z^2\nabla_\rho^2$ and $r = \rho/Z$,

$$\hat{H}(Z) = Z^2 \left(-\frac{1}{2}\nabla_\rho^2 - \frac{1}{\rho} \right).$$

Thus, for a consistently scaled Gaussian basis, the variational energy satisfies

$$E_{\mathrm{G}}(Z) = Z^2 E_{\mathrm{G}}(1). \tag{2.20}$$

Because the exact energy also scales as Z^2, the relative energy error is unchanged under this transformation.

The normalised wavefunction scales as $\psi_Z(r) = Z^{3/2}\psi_1(Zr)$, giving the corresponding radial probability relation,

$$P_Z(r) = ZP_1(Zr), \qquad \frac{P_Z(\rho/Z)}{Z} = P_1(\rho). \tag{2.21}$$

The scaled radial distributions should therefore coincide when expressed in terms of $\rho = Zr$. Together, these relations provide specific theoretical predictions for the nuclear-charge calculations presented in Chapter 4.

Chapter 3

Computational Methodology

The calculations examine how the accuracy of a Gaussian approximation depends on basis size, exponent selection and nuclear charge. The exact hydrogenic solution provides a reference against which energy and wavefunction errors can be measured, while independent numerical checks distinguish limitations of the basis from errors in the computation. The methods developed from Chapter 2 are also used to investigate near-linear dependence and the transferability of a hydrogen-optimised basis to more strongly bound one-electron ions.

3.1 Variational Calculation

The calculations were implemented in Python using NumPy for array operations, SciPy for numerical linear algebra and integration, and Matplotlib for plotting. Atomic units are used throughout. For specified values of Z, N, α and β, the even-tempered Gaussian exponents are generated using Eq. (2.5).

The overlap and Hamiltonian matrices are constructed from the analytical expressions in Eqs. (2.8)–(2.11), derived in Appendix A. Evaluating these matrix elements analytically avoids numerical integration errors and allows the calculation to be repeated efficiently for different basis parameters.

The expansion coefficients are obtained by solving the generalised eigenvalue problem in Eq. (2.7) using `scipy.linalg.eigh(H, S)`. The lowest eigenvalue gives the variational ground-state energy, while the corresponding eigenvector provides the Gaussian expansion coefficients. The coefficients are normalised using the overlap matrix, and their overall sign is chosen so that the reconstructed ground-state wavefunction is positive near the nucleus.

The reference hydrogen calculation uses

$$Z = 1, \qquad N = 12, \qquad \alpha = 0.01 \text{ bohr}^{-2}, \qquad \beta = 1.9. \tag{3.1}$$

These parameters provide a reproducible starting point rather than an optimised basis. The resulting energy, wavefunction and radial probability distribution are compared with the exact expressions in Eqs. (2.2) and (2.3).

The mean radius is calculated directly from the Gaussian coefficients and analytical radial matrix elements,

$$\langle r \rangle_{\mathrm{G}} = \sum_{i,j} c_i c_j \frac{2\pi A_i A_j}{(\lambda_i + \lambda_j)^2}. \tag{3.2}$$

The wavefunction error defined in Eq. (2.14) is evaluated separately using numerical radial integration. This provides a comparison of both the ground-state energy and the spatial accuracy of the reconstructed wavefunction.

3.2 Investigating Accuracy and Basis Limitations

The first calculations examine how energy accuracy changes as the number of Gaussian functions increases. For an even-tempered basis, increasing N at fixed α and β adds progressively narrower functions without extending the diffuse end of the basis. The rate of convergence therefore depends on the exponent distribution as well as the number of functions. The convergence calculations use $N = 2, \ldots, 30$, with $\alpha = 0.01$ bohr^{-2} and $\beta = 1.6$, 1.9 and 2.3. Both the relative energy error and overlap condition number $\kappa(\mathbf{S})$ are recorded. Comparing these quantities allows improvements in variational accuracy to be considered alongside increasing numerical sensitivity as the basis grows.

Exponent selection is examined separately using a 61×61 parameter map at $N = 12$. The map covers $10^{-3} \leq \alpha \leq 10^{-1}$ bohr^{-2} and $1.35 \leq \beta \leq 3.05$. This shows how the energy error depends jointly on the most diffuse Gaussian and the spacing between successive exponents, rather than on basis size alone.

A coarser basis-selection search is then performed over $N = 2, \ldots, 20$, using 13 logarithmically spaced values of α between 10^{-3} and 10^{-1} bohr^{-2}, and 15 linearly spaced values of β between 1.4 and 2.8. For each prescribed energy-error tolerance, the smallest tested basis size containing at least one acceptable parameter combination is recorded. The search proceeds in increasing N and stops once all three energy-error tolerances have been met. In the final calculation, all parameter combinations for $N = 2, \ldots, 8$ were evaluated, giving 1365 attempted calculations, with no eigensolver failures. Only candidates satisfying $\kappa(\mathbf{S}) \leq 10^{12}$ are accepted in the basis-selection search. This limits the inclusion of poorly conditioned solutions but does not guarantee numerical accuracy. The reported bases are therefore the smallest successful choices within the tested grid, rather than globally optimal solutions. The separate convergence calculations and dense parameter map are not subject to this cutoff, allowing successfully solved cases with larger condition numbers to be examined.

Energy convergence is then compared with wavefunction convergence. As discussed in Section 2, a small variational energy error does not necessarily imply that the approximate wavefunction is equally accurate. The L^2 error in Eq. (2.14) is therefore evaluated independently by numerically integrating the squared difference between the Gaussian and exact wavefunctions. The integration is divided into separate radial intervals to resolve both the region near the nucleus and the decaying tail. Integrating the squared difference directly also avoids subtracting nearly equal overlap terms, which can reduce numerical precision when the wavefunction approximation is highly accurate.

The electron–nucleus cusp provides a separate test of accuracy near the nucleus. A finite Gaussian expansion cannot reproduce the exact derivative in Eq. (2.16). The local energy defined in Eq. (2.17) is therefore evaluated as a function of radius to examine how this limitation affects the approximation near the singular Coulomb potential. The Gaussian and exact wavefunctions are also compared directly near the origin.

Near-linear dependence is investigated using a separate Gaussian basis with closely spaced exponents, $\beta = 1.3$ and $N = 4, \ldots, 40$. Canonical orthogonalisation is applied to the overlap matrix to identify and remove nearly dependent combinations of basis functions. Three relative overlap-eigenvalue thresholds, $\tau = 10^{-6}$, 10^{-9} and 10^{-12}, are tested to examine how the amount of pruning affects the calculated energy.

The retained basis is re-orthogonalised to reduce floating-point errors associated with small overlap eigenvalues. The Hamiltonian is then transformed into the reduced basis, and its lowest eigenvalue is used to calculate the relative energy error. Numerical reliability checks are applied to the transformed basis and calculated eigenpairs before accepting the results.

For each threshold and basis size, the number of retained basis directions and the relative energy error are recorded. These quantities allow the reduction in basis dimension to be considered alongside any loss of variational accuracy caused by removing basis directions.

The final calculations examine whether a Gaussian basis chosen for hydrogen remains suitable at larger nuclear charges. For $Z = 1, \ldots, 25$, the original hydrogen exponents are compared with those scaled according to Eq. (2.19),

$$\lambda_i(Z) = Z^2 \lambda_i(1).$$

The exact hydrogenic orbital contracts on a length scale proportional to $1/Z$, so fixed hydrogen exponents are expected to become increasingly diffuse relative to the electron distribution. The $Z = 25$ ion, Mn^{24+}, provides a detailed example of this effect. The radial probability distributions are also compared in the scaled coordinate $\rho = Zr$, using $P_Z(\rho/Z)/Z$. This tests the radial scaling relation in Eq. (2.21) and allows the spatial results to be compared with the energy calculations.

3.3 Numerical Verification and Stability

Numerical verification was performed to distinguish errors in the computational implementation from limitations of the Gaussian approximation itself. Since an accurately solved eigenvalue problem does not necessarily imply that its underlying matrices are correct, the calculations were assessed through complementary consistency checks and independent numerical comparisons.

The overlap and Hamiltonian matrices were checked for symmetry, while the diagonal overlap elements were verified to equal unity, consistent with the normalisation of the individual Gaussian functions. The resulting eigenpairs were also tested for wavefunction normalisation, agreement between the calculated eigenvalue and Rayleigh quotient, and satisfaction of the variational energy bound.

To assess how accurately the computed eigenpairs satisfy the generalised eigenvalue equation, the relative residual was calculated as

$$\eta = \frac{\|\mathbf{Hc} - E\mathbf{Sc}\|_2}{\|\mathbf{c}\|_2 \left(\|\mathbf{H}\|_2 + |E|\|\mathbf{S}\|_2\right)}. \tag{3.3}$$

Normalising by the matrix and eigenvalue scales permits meaningful comparisons between calculations with different basis parameters and nuclear charges. However, a small residual only establishes consistency with the constructed matrices. To verify the matrix elements independently, selected overlap, kinetic-energy and Coulomb integrals were compared against direct numerical radial integration.

The wavefunction L^2 error was independently recalculated using a fine radial grid to assess the reliability of the numerical integration. The expected hydrogenic energy scaling and invariance of the overlap condition number were also tested for consistently scaled exponents at $Z = 1, 2, 5, 10, 25, 100$. The $Z = 100$ case served exclusively as an additional numerical scaling check and was not included in the physical transferability analysis.

Additional checks were applied to the basis-pruning calculations, where small overlap eigenvalues can increase sensitivity to floating-point errors. Following re-orthogonalisation, the retained basis was checked for orthonormality, the reduced Hamiltonian for symmetry, and the resulting eigenpairs for residual accuracy and compliance with the variational bound. All tested basis sizes satisfied these checks for each of the three pruning thresholds.

The numerical discrepancies obtained from the final calculations are summarised in Table 3.1.

Table 3.1: Numerical verification results. The matrix-element discrepancy is the maximum relative difference across 12 independent comparisons. The energy-scaling discrepancy is the maximum absolute difference between $E(Z)/Z^2$ and $E(1)$.

Verification test	Measured discrepancy
Relative eigenpair residual (reference basis)	2.404×10^{-16}
Matrix elements (12 comparisons)	6.565×10^{-13}
Z^2 energy scaling ($Z = 1, 2, 5, 10, 25, 100$)	3.44×10^{-15} Ha
Independent L^2 integration ($N = 30$)	2.127×10^{-10}

The L^2 discrepancy represents the relative difference between the two integration methods, while the other dimensionless entries are relative numerical discrepancies. All implemented verification checks passed, supporting the internal consistency of the calculations and the reliability of the reported numerical results.

These checks do not, however, establish the physical accuracy of a finite Gaussian expansion. In particular, numerical convergence of the eigenvalue problem cannot eliminate the limitations associated with the electron–nucleus cusp or an incomplete basis. Likewise, the condition-number cutoff used during basis selection restricts numerical sensitivity without guaranteeing a particular level of accuracy. The distinction between computational reliability and the accuracy of the physical approximation is examined in Chapter 4.

Chapter 4

Results and Discussion

4.1 Hydrogen Benchmark

The reference hydrogen calculation provides an initial assessment of how accurately the Gaussian expansion reproduces the exact ground state. Using the parameters in Eq. (3.1), the calculated energy is -0.4995891522 Hartree, compared with the exact value of -0.5 Hartree. The relative energy error is 8.2170×10^{-4}, or approximately 0.0822%. The calculated energy lies above the exact value, as required by the variational principle. The calculated mean radius is 1.50124491 bohr, compared with the exact value of 1.5 bohr. This corresponds to a relative radius error of approximately 8.30×10^{-4}, or 0.0830%. The close agreement in both quantities suggests that the reference basis represents the overall ground-state distribution well, although neither measure establishes its accuracy at every radius.

Table 4.1 summarises the benchmark results. The overlap condition number of 8.3295×10^4 also provides a reference for the conditioning behaviour examined later.

Table 4.1: Hydrogen ground-state benchmark for $N = 12$, $\alpha = 0.01\,\mathrm{bohr}^{-2}$ and $\beta = 1.9$. Exact reference values are obtained from the analytical hydrogenic ground-state solution in Section 2.1 for $Z = 1$.

Quantity	Exact	Gaussian basis
Energy (Hartree)	-0.500000	-0.499589
Mean radius (bohr)	1.500000	1.501245
Relative energy error	0	8.2170×10^{-4}
Overlap condition number	—	8.3295×10^4

Figure 4.1 compares the exact and approximate wavefunctions directly. Their overall shapes agree closely, but the signed difference reveals a more noticeable discrepancy near the nucleus. At $r = 0$, the Gaussian expansion gives $\psi_{\mathrm{G}}(0) = 0.516211566\ \mathrm{bohr}^{-3/2}$, compared with $\psi_{1s}(0) = 0.564189584\ \mathrm{bohr}^{-3/2}$. The relative amplitude difference is:

$$\frac{|\psi_{\mathrm{G}}(0) - \psi_{1s}(0)|}{|\psi_{1s}(0)|} \approx 0.0850,$$

corresponding to an underestimate of approximately 8.5%.

The electron density depends on the square of the wavefunction amplitude. The corresponding relative density difference at the nucleus is:

$$1 - \frac{|\psi_{\mathrm{G}}(0)|^2}{|\psi_{1s}(0)|^2} \approx 0.163,$$

or approximately 16.3%. These local errors are substantially larger than the 0.0822% energy error. The variational energy therefore gives a favourable assessment of the reference basis that is not representative of its accuracy at the nucleus. This matters for physical quantities that depend strongly on the electron density or wavefunction behaviour in the near-nuclear region.

The radial probability distributions in Fig. 4.2 show even closer agreement. The radial probability density contains the factor $4\pi r^2$, which suppresses the contribution from the immediate vicinity of the nucleus. Consequently, the distributions can appear nearly identical despite the appreciable difference between the wavefunction amplitudes at $r = 0$. This also helps explain why the mean radius is reproduced accurately. The expectation value $\langle r \rangle$ depends on the entire radial distribution, with additional weighting towards larger radii, rather than on the wavefunction at a single point. The benchmark therefore reveals two aspects of the approximation. The Gaussian expansion captures the overall ground-state energy and radial distribution well, but its local accuracy

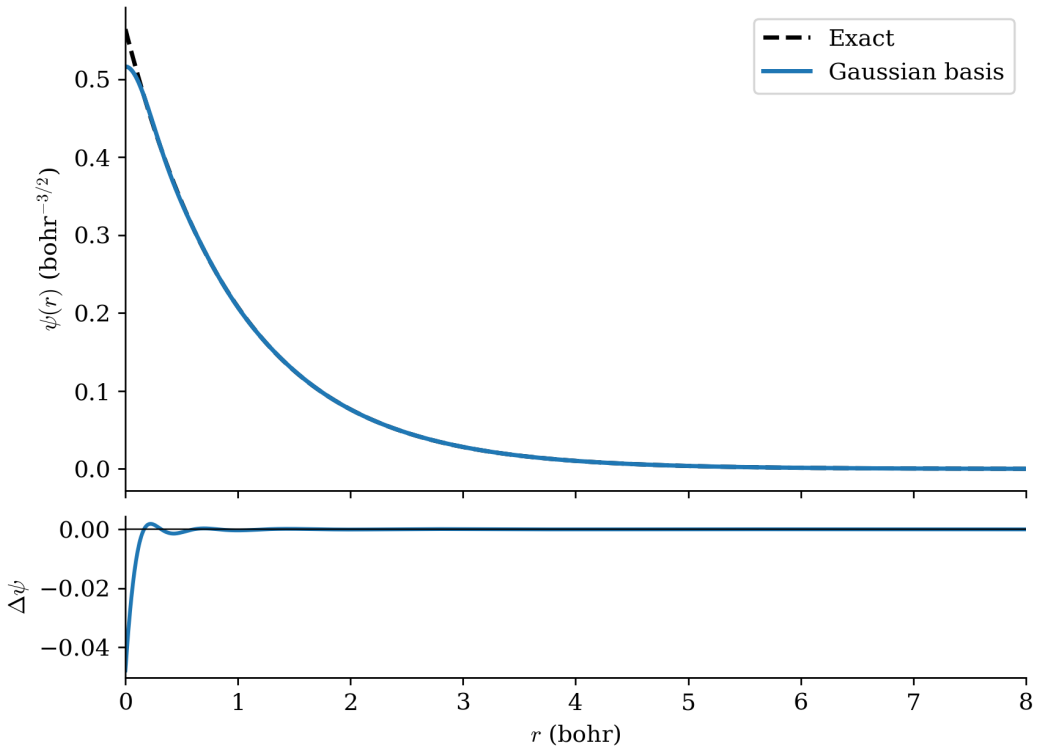


Figure 4.1: Exact and Gaussian hydrogen ground-state wavefunctions, together with their signed difference. The discrepancy is most apparent near the nucleus, despite close agreement over most of the radial range.

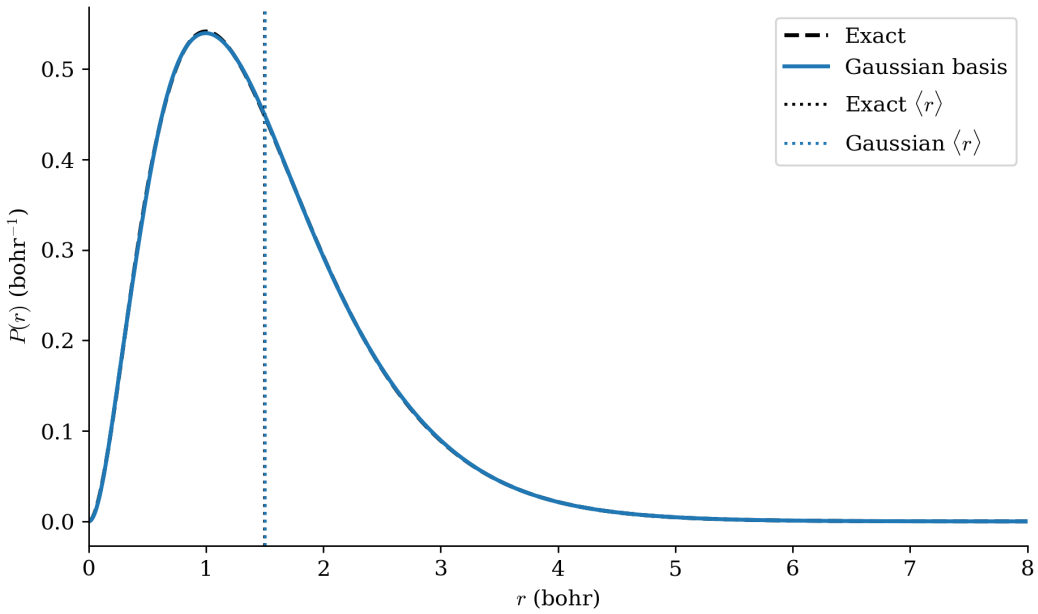


Figure 4.2: Exact and Gaussian radial probability distributions for hydrogen, with their mean radii indicated. The mean radius is distinct from the position of maximum radial probability.

is considerably poorer near the nucleus. This difference motivates the separate convergence and near-nuclear tests that follow.

4.2 Basis-Size Convergence and Exponent Selection

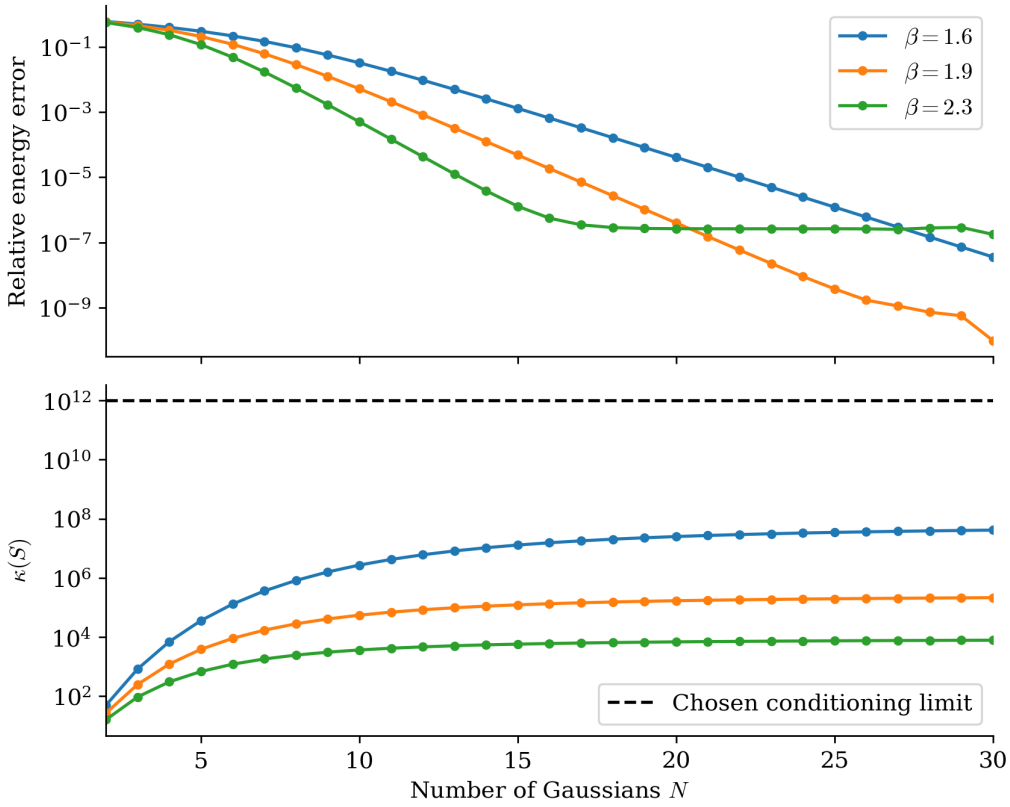


Figure 4.3: Relative hydrogen energy error and overlap-matrix condition number against basis size for three exponent ratios. The dashed condition-number threshold is the screening criterion used in the separate basis-selection search, rather than a universal stability limit.

Figure 4.3 shows that the rate of energy convergence depends strongly on the spacing between Gaussian exponents. With $\alpha = 0.01$ bohr^{-2} held fixed, the three values of β produce different convergence behaviour as N increases. For the tested range, $\beta = 2.3$ initially reduces the energy error more rapidly than the smaller exponent ratios. However, its improvement subsequently slows and the curve approaches a plateau. The $\beta = 1.9$ sequence continues improving over a wider range of basis sizes. This difference is important because it shows that the most effective exponent distribution depends on the number of functions available. A ratio that gives rapid improvement for a small basis does not necessarily provide the best approximation as the basis becomes larger.

The condition-number results provide a separate view of numerical behaviour. The overlap condition number generally increases with basis size, particularly when successive exponents are closely spaced. This is expected because Gaussians with similar widths overlap strongly, making some combinations of functions difficult to distinguish numerically. However, the energy plateau for $\beta = 2.3$ occurs well below the chosen condition-number threshold. The available evidence therefore does not establish numerical instability as the cause of the plateau.

A possible explanation follows from the structure of the even-tempered sequence. At fixed α and β, increasing N introduces progressively narrower Gaussians without adding more diffuse functions. Once the short-range behaviour is adequately represented, further functions may provide little improvement if the remaining error lies in radial regions that are not well covered. This interpretation is consistent with the sensitivity to exponent selection examined below. However, the convergence curves alone do not isolate the source of the remaining error, so the plateau cannot be assigned a unique cause.

Figure 4.4 examines the combined effect of α and β at fixed basis size. The variation in energy error shows that

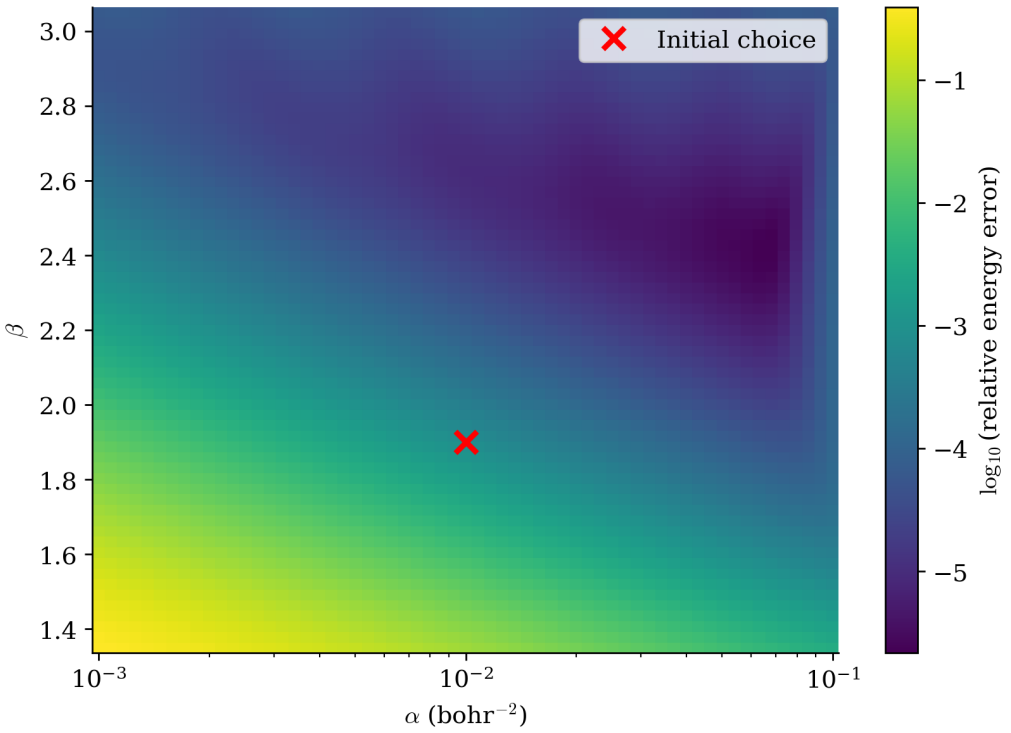


Figure 4.4: Relative hydrogen energy error across the tested even-tempered exponent parameters at $N = 12$. The red cross marks the reference parameters $\alpha = 0.01$ bohr^{-2} and $\beta = 1.9$, not an optimised solution.

the reference parameters are not uniquely favourable and that the distribution of exponents has a substantial influence on the quality of the approximation. The smallest exponent α determines the broadest Gaussian, while β determines how rapidly the sequence reaches narrower functions. Their effects are connected because the largest exponent is $\lambda_N = \alpha\beta^{N-1}$. Changing α shifts the overall exponent range, whereas changing β alters both the spacing and the extent of that range. Consequently, the parameters cannot be assessed independently when judging the radial coverage of a finite basis. The parameter map therefore provides a physical explanation for the different convergence curves. Adding functions is useful only when their widths improve the representation of the wavefunction. A smaller basis with appropriately chosen exponents can outperform a larger basis whose exponent distribution is less suitable.

To quantify this effect, a discrete parameter search was used to identify the smallest tested basis satisfying selected relative energy-error targets. The results are given in Table 4.2.

Table 4.2: Smallest tested Gaussian bases satisfying selected relative energy-error targets. Results are restricted to the discrete parameter grid described in Chapter 3, with $\kappa(\mathbf{S}) \leq 10^{12}$.

Target error	N	α (bohr^{-2})	β	Actual error
10^{-2}	5	0.100000	2.8	2.5755×10^{-3}
10^{-3}	6	0.100000	2.8	6.2944×10^{-4}
10^{-4}	8	0.068129	2.8	6.6406×10^{-5}

The search evaluated 1,365 parameter combinations before its early-stopping condition was reached. All attempted generalised eigenvalue solves completed successfully. For the least restrictive target, a relative energy error below 10^{-2} was achieved with five Gaussians. Six functions were sufficient for an error below 10^{-3}, while the strictest target of 10^{-4} was reached with eight functions.

The eight-function basis uses $\alpha = 0.068129$ bohr^{-2} and $\beta = 2.8$, giving a relative energy error of 6.6406×10^{-5}. This is a substantial improvement over the twelve-function reference basis, whose relative energy error is 8.2170×10^{-4}. The selected basis achieves an error approximately 12.4 times smaller using four fewer functions. The result demonstrates that increasing basis size is not necessarily the most efficient way to improve accuracy. Selecting exponents that better represent the relevant radial length scales can be more effective than adding

functions to a fixed sequence.

An important qualification is that all three selected values of β lie at the upper boundary of the tested search range, $\beta = 2.8$. Two of the selected α values also lie at their upper boundary. These boundary selections suggest that the chosen parameter ranges may influence the reported results. Extending the search beyond these limits or using a finer grid could identify other successful combinations, potentially with fewer functions. The results therefore establish the smallest successful basis sizes within the tested discrete grid, not globally optimal Gaussian bases. Nevertheless, the comparison with the reference calculation provides clear evidence that exponent selection is central to variational accuracy.

4.3 Wavefunction Accuracy and Near-Nuclear Behaviour

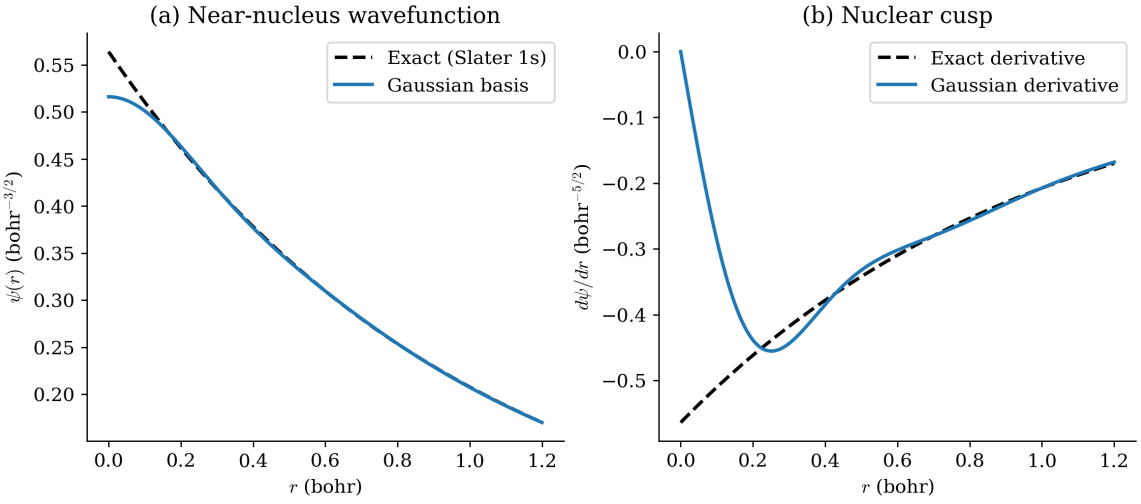


Figure 4.5: Exact and Gaussian hydrogen wavefunctions and radial derivatives near the nucleus. The Gaussian derivative vanishes at the origin, whereas the exact derivative satisfies the electron–nucleus cusp condition.

Figure 4.5 examines the near-nuclear region more closely, where the reference wavefunction shows its largest visible discrepancy. For hydrogen, the exact radial derivative at the origin is

$$\psi'_{1s}(0) = -\frac{1}{\sqrt{\pi}}\ \text{bohr}^{-5/2}.$$

By contrast, the derivative of the finite Gaussian expansion is zero.

This difference follows from the functional form of the basis rather than the numerical solution. Every centred spherical Gaussian has zero radial derivative at the origin, so any finite linear combination also has zero derivative there. The exact hydrogenic wavefunction instead satisfies the non-zero electron–nucleus cusp condition in Eq. (2.16) [9, 10]. Increasing the number of Gaussians or changing their coefficients can improve the approximation over a small but finite region near the nucleus. However, no finite combination of these functions can reproduce the exact derivative at $r = 0$. This is an inherent limitation of the finite spherical Gaussian basis and cannot be removed simply by solving the variational problem more accurately.

The local-energy comparison in Fig. 4.6 shows a further consequence of the missing cusp. For an exact eigenstate, the local energy defined in Eq. (2.17) equals the corresponding eigenvalue wherever the wavefunction is non-zero. For hydrogen, this is the constant value -0.5 Hartree.

The Gaussian approximation behaves differently. For a finite spherical Gaussian expansion, the kinetic-energy contribution remains finite as $r \to 0$, whereas the Coulomb potential contains the singular term $-Z/r$. Because the approximate wavefunction is non-zero at the nucleus, its local energy diverges negatively as $r \to 0^+$. The difference is directly related to the cusp condition. In the exact state, the non-zero radial derivative produces a singular kinetic contribution that cancels the Coulomb singularity. The finite Gaussian expansion cannot provide the same cancellation.

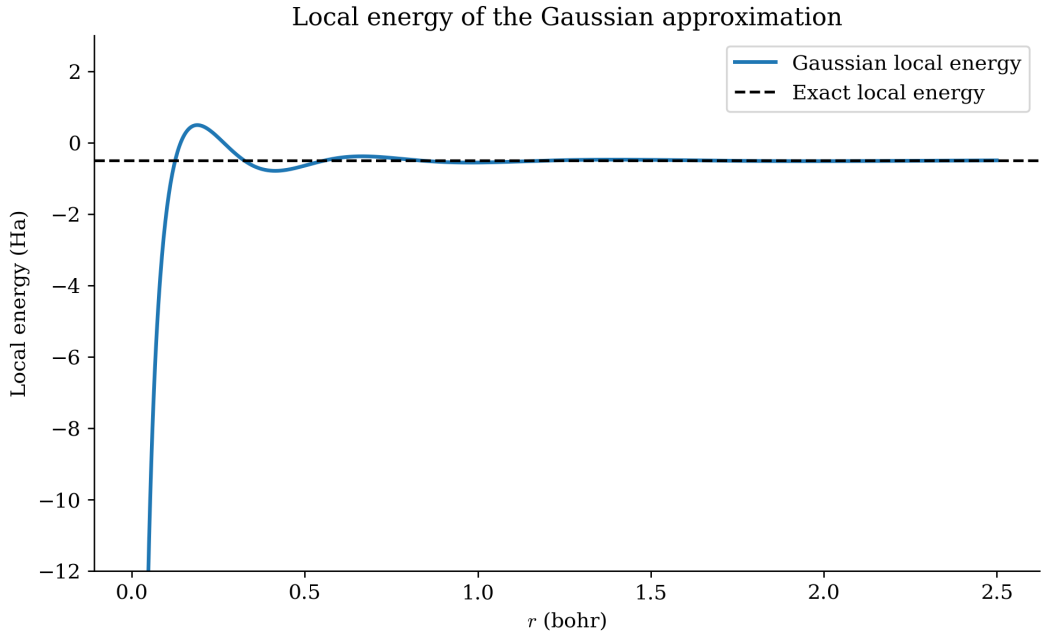


Figure 4.6: Local energy of the Gaussian hydrogen approximation compared with the constant exact ground-state energy. The strong near-nuclear variation results from the missing cusp.

Importantly, this local divergence does not make the total variational energy infinite. The energy expectation value is integrated over three-dimensional space, with radial volume element $4\pi r^2\,\mathrm{d}r$. Near the nucleus, the Coulomb contribution therefore behaves as an integrable term proportional to $r\,\mathrm{d}r$ for a finite, non-zero wavefunction amplitude. The total energy remains finite even though the local energy diverges. This explains how a Gaussian basis can give an accurate integrated energy while retaining a systematic error in its behaviour at the nucleus.

Figure 4.7 compares the relative energy error with the three-dimensional L^2 wavefunction error as the basis size increases. Both quantities generally decrease with increasing N, but their rates of convergence are different. The energy becomes highly accurate before the wavefunction error has decreased by a comparable amount. For a normalised approximate ground state, Eq. (2.15) expresses the variational energy error as a quadratic, energy-weighted function of the wavefunction error, whereas the L^2 norm measures the wavefunction difference directly. The energy can therefore improve substantially even when a small spatial error remains. This explains why energy convergence alone is insufficient to establish the accuracy of the reconstructed wavefunction, although it does not identify the cause of the L^2-error plateau discussed below.

The wavefunction L^2 error begins to plateau at approximately $N = 20$, remaining near 1.5×10^{-5} despite continued improvement in the variational energy. At $N = 30$, the calculated wavefunction error is

$$\varepsilon_\psi \approx 1.50493050 \times 10^{-5}.$$

Adaptive numerical integration and fine-grid integration agree to a relative difference of 2.127×10^{-10}. The close agreement indicates that the observed wavefunction error is not simply an inconsistency between the two integration methods. It also supports the reliability of the numerical evaluation of the L^2 norm.

However, the calculations do not establish the cause of the apparent wavefunction-error plateau. Possible influences include the distribution of Gaussian exponents, the remaining limitations of the finite basis and numerical sensitivity at larger N. The present results do not distinguish between these possibilities. The plateau should therefore not be interpreted as a fundamental accuracy limit of Gaussian basis expansions. Further calculations with different exponent ranges, larger bases or higher numerical precision would be needed to identify its origin.

The canonical orthogonalisation results in Fig. 4.8 examine the effect of removing nearly dependent combinations of Gaussian functions. As the exponent ratio approaches unity, neighbouring Gaussians become increasingly similar, producing small eigenvalues of the overlap matrix. Pruning these directions reduces the effective dimension of the variational space, but can also remove combinations that contribute to the accuracy of the ground state.

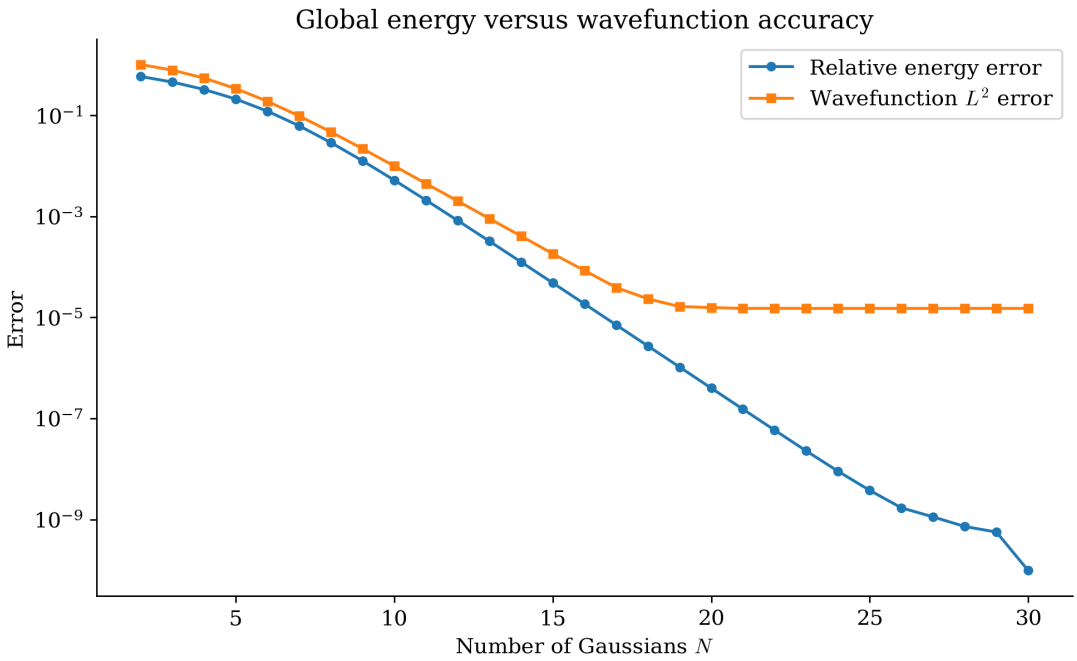


Figure 4.7: Convergence of the relative variational energy error and the three-dimensional L^2 wavefunction error with increasing Gaussian basis size. The two quantities converge at different rates.

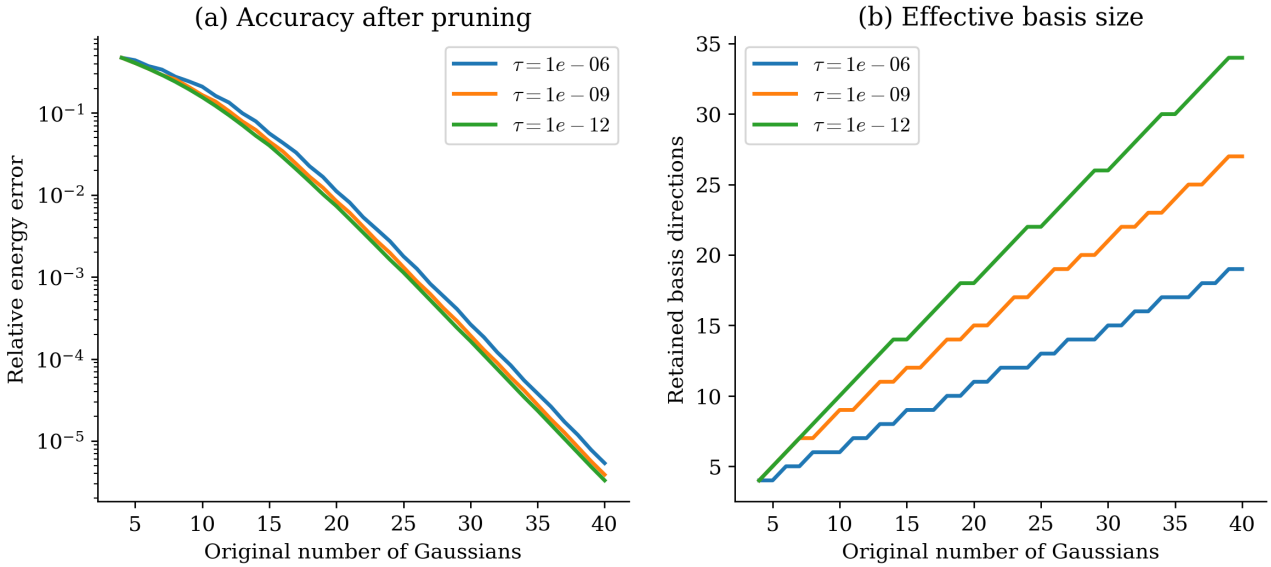


Figure 4.8: Canonical orthogonalisation test showing (a) relative ground-state energy error and (b) retained basis dimension for three overlap-eigenvalue thresholds. Following re-orthogonalisation, all tested basis sizes passed the numerical reliability checks, giving continuous energy-error curves.

All 111 pruning calculations passed the implemented reliability checks following re-orthogonalisation, with no exclusions at any of the three thresholds. The energy-error curves consequently extend across the complete tested range, $N = 4, \ldots, 40$. No energy-error values are omitted from the final plot. At $N = 40$, the thresholds 10^{-6}, 10^{-9} and 10^{-12} retain 19, 27 and 34 basis directions, respectively. The most aggressive threshold therefore removes 21 of the original 40 directions, a reduction of 52.5%. The intermediate and least aggressive thresholds remove 32.5% and 15.0%, respectively. These reductions show how strongly the closely spaced Gaussians overlap: many nominal basis functions add little independent variational freedom. The energy-error curves reveal the corresponding trade-off within this experiment, with smaller thresholds generally giving lower errors and larger thresholds producing more compact bases.

The results demonstrate that nominal basis size can substantially overstate the number of distinct directions needed for a given level of energy accuracy. Canonical orthogonalisation offers a practical way to control near-linear dependence, provided that the retained dimension, variational error and numerical checks are assessed together. Passing those checks establishes the reliability of the reported calculations under the chosen criteria, not a universal guarantee of stability for arbitrary basis parameters.

4.4 Transferability and Nuclear-Charge Scaling

The final calculations examine whether a Gaussian basis constructed for hydrogen can be reused for more strongly bound one-electron ions. The exact hydrogenic ground state contracts on a characteristic length scale proportional to $1/Z$. Consequently, an unchanged hydrogen exponent sequence becomes increasingly broad relative to the physical electron distribution as the nuclear charge increases.

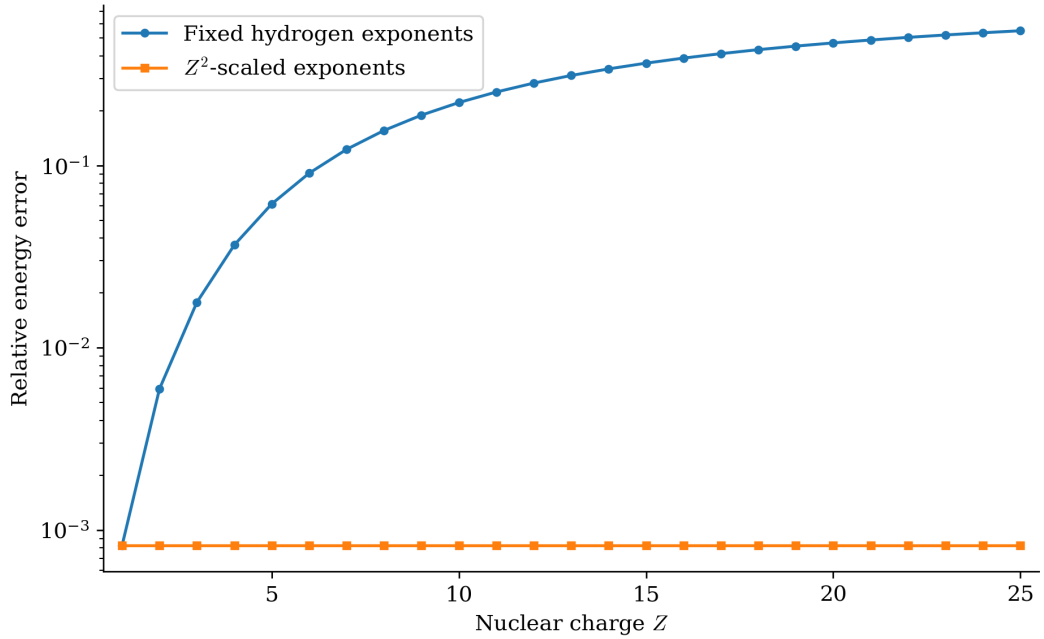


Figure 4.9: Relative ground-state energy error against nuclear charge for fixed hydrogen exponents and exponents scaled by Z^2. The scaled basis preserves the hydrogen relative energy error.

Figure 4.9 compares the relative energy errors obtained with fixed and Z^2-scaled exponents. For the unchanged hydrogen basis, the relative error increases from 8.2170×10^{-4} at $Z = 1$ to approximately 0.549 at $Z = 25$. The fixed basis therefore becomes progressively less effective at representing the more strongly bound ground states. By contrast, the Z^2-scaled basis retains the hydrogen relative energy error throughout the tested range. This agrees with Eq. (2.20), which predicts:

$$E_{\mathrm{G}}(Z) = Z^2 E_{\mathrm{G}}(1),$$

when the Gaussian exponents are scaled consistently. Since the exact hydrogenic energy also scales as Z^2, the relative energy error remains unchanged.

Although the scaled basis maintains a constant relative energy error, this result applies within the non-relativistic hydrogenic model. As Z increases, the characteristic electron velocity scales approximately as $v/c \sim Z\alpha_{\mathrm{fs}}$, where $\alpha_{\mathrm{fs}} \approx 1/137$ is the fine-structure constant. Relativistic corrections therefore become increasingly important at higher nuclear charges and are not included in the present calculations.

Table 4.3: Exact and variational ground-state energies for hydrogen and hydrogenic manganese (Mn^{24+}). Exact values follow from $E_{1s} = -Z^2/2$ (Section 2.1); the fixed- and scaled-exponent results are calculated as described in Section 3.2. Energies are in Hartree.

System	Exact	Fixed exponents	Scaled exponents
H ($Z = 1$)	-0.500000	-0.499589	-0.499589
Mn^{24+} ($Z = 25$)	-312.500000	-140.956593	-312.243220

The $Z = 25$ energies in Table 4.3 show the magnitude of the difference. For hydrogenic Mn^{24+}, the exact ground-state energy is -312.5 Hartree. Using the unchanged hydrogen basis gives -140.95659344 Hartree, corresponding to a relative energy error of approximately 54.9%. With Z^2-scaled exponents, the calculated energy becomes -312.24322013 Hartree, reducing the relative error to approximately 0.0822%. Compared with the fixed basis, this is a reduction in relative energy error by a factor of approximately 668, without increasing the number of Gaussian functions. The comparison demonstrates that matching the exponent distribution to the physical length scale can be more effective than simply enlarging a poorly matched basis.

The improvement follows directly from the coordinate transformation $\rho = Zr$. In the scaled coordinate, the Gaussian exponent appears through the combination λ_i/Z^2. Multiplying each exponent by Z^2 therefore preserves the original distribution of Gaussian widths relative to the contracted orbital. The exponent scaling thus adapts the existing basis to the contracted orbital, rather than introducing additional functions.

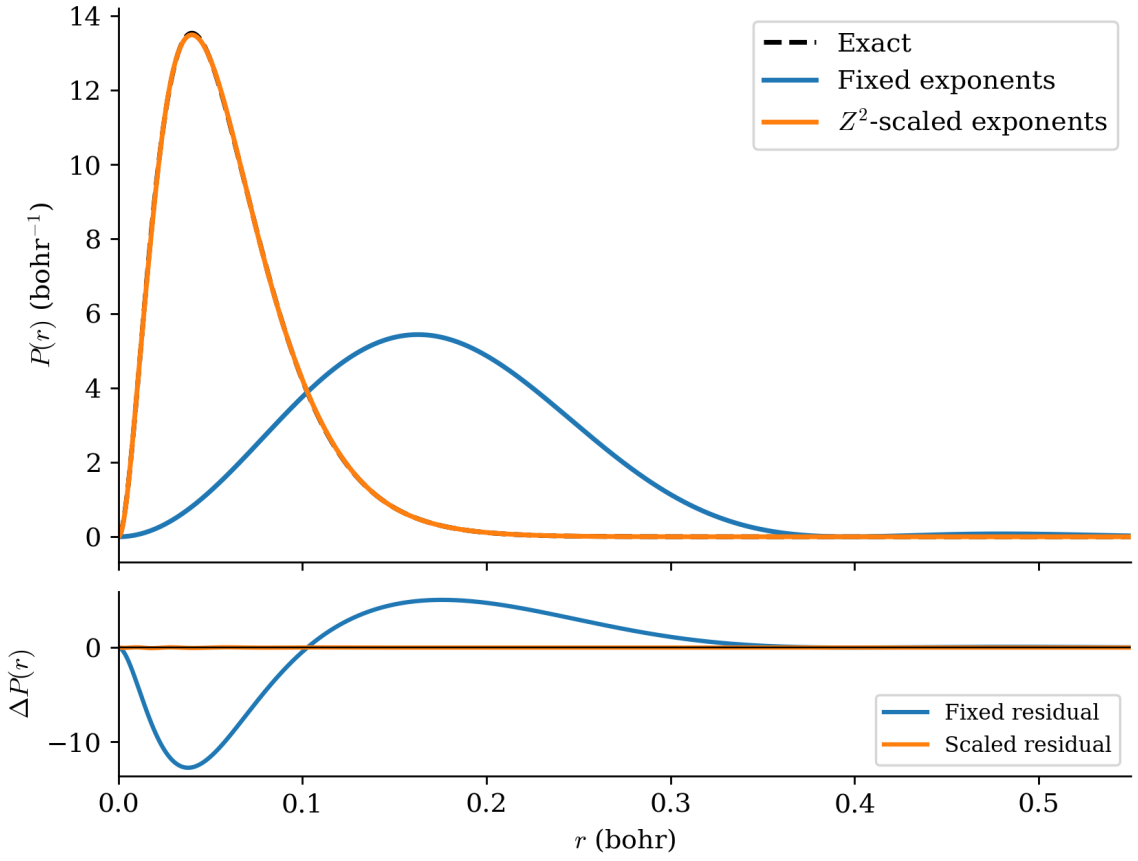


Figure 4.10: Exact, fixed-basis and Z^2-scaled radial probability distributions for hydrogenic Mn^{24+}, including residuals relative to the exact distribution.

The radial probability distributions in Fig. 4.10 show why the fixed basis performs poorly. The exact Mn^{24+} ground state is strongly concentrated near the nucleus. The unchanged hydrogen basis places too much probability

at larger radii and does not reproduce the required contraction. The scaled basis instead follows the exact radial distribution much more closely. The improvement is therefore not confined to the variational energy. Scaling the exponents also gives a substantially better description of the spatial electron distribution. This comparison is important because it connects the large energy error of the fixed basis to a physically identifiable problem: its Gaussian widths are poorly matched to the size of the orbital.

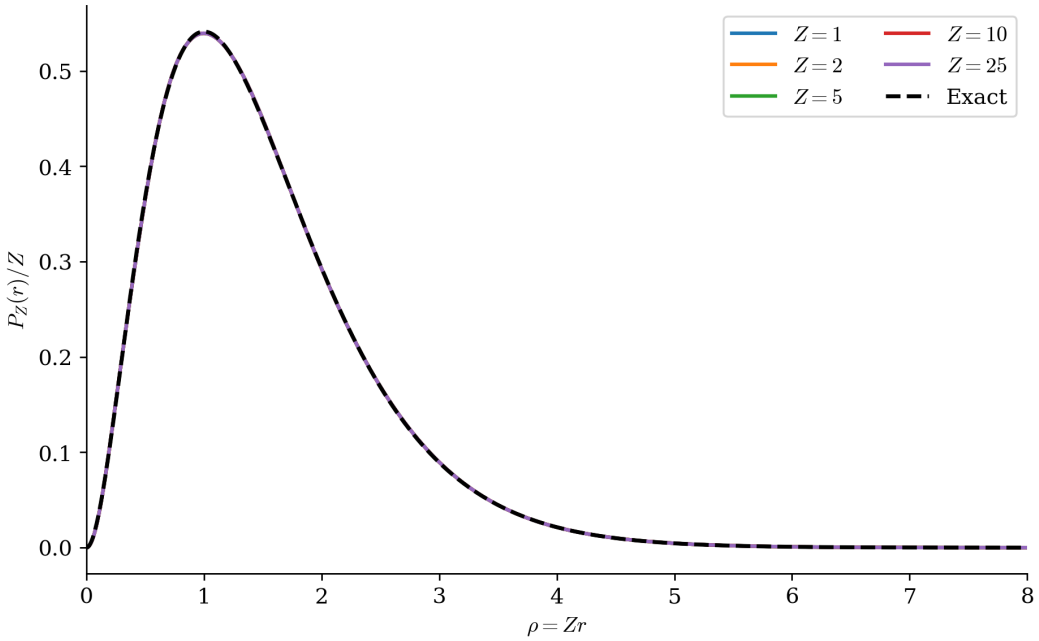


Figure 4.11: Scaled radial probability $P_Z(\rho/Z)/Z$ against $\rho = Zr$ for several nuclear charges. The overlapping curves verify the predicted hydrogenic coordinate scaling.

Figure 4.11 provides an additional check of the scaling relation. Because $P_Z(r)\,\mathrm{d}r$ represents a probability, the coordinate transformation $\rho = Zr$ requires

$$P_Z(r)\,\mathrm{d}r = \frac{P_Z(\rho/Z)}{Z}\,\mathrm{d}\rho.$$

The scaled radial distributions coincide when expressed in terms of ρ, as predicted by Eq. (2.21). This confirms that the numerical implementation reproduces the known coordinate scaling of the non-relativistic hydrogenic Hamiltonian. The collapse is therefore a verification of the theoretical relation rather than evidence for a new scaling law. It also provides a useful check that the energy agreement is accompanied by the expected transformation of the wavefunction's spatial distribution.

Taken together, the calculations show that the accuracy of a Gaussian basis depends on its exponent distribution, the number of functions and the physical quantity being approximated. Within the tested parameter grid, eight carefully selected Gaussians were sufficient to achieve a relative hydrogen energy error below 10^{-4}, outperforming the twelve-function reference basis. However, the near-nuclear wavefunction discrepancy and missing cusp demonstrate that accurate total energies do not guarantee equally accurate local properties. The conditioning results also show why numerical reliability must be assessed alongside variational accuracy, particularly as the basis functions become nearly linearly dependent.

Finally, the nuclear-charge calculations demonstrate that a basis must represent the appropriate physical length scale. For the fixed-nucleus, non-relativistic one-electron Hamiltonian, this adjustment follows exactly from the Z^2 scaling of Gaussian exponents. A useful Gaussian basis must therefore balance exponent coverage, variational accuracy and numerical reliability, rather than being judged by basis size alone.

Chapter 5

Conclusion and Limitations

5.1 Main Findings

The accuracy of a variational Gaussian expansion depends not only on the number of basis functions, but also on their radial extent, exponent spacing and numerical conditioning. By comparing even-tempered Gaussian expansions with exact hydrogenic ground states, the calculations established how these factors affect energy convergence, wavefunction accuracy and the transferability of a basis between nuclear charges.

For the twelve-function hydrogen reference basis, the calculated ground-state energy was -0.499589 Hartree, corresponding to a relative error of 8.2170×10^{-4} (0.0822%). The mean radius was 1.501245 bohr, compared with the exact value of 1.5 bohr, and the radial probability distribution closely reproduced the exact result. These comparisons demonstrate that a relatively small Gaussian basis can describe the overall hydrogen ground state accurately. However, the agreement was not uniform across all physical quantities. The wavefunction amplitude at the nucleus was underestimated by approximately 8.5%, corresponding to a 16.3% underestimate of the electron density. Thus, the small variational energy error did not imply equally accurate local wavefunction behaviour. This distinction is particularly important for quantities sensitive to the near-nuclear electron distribution.

The basis-size calculations showed that convergence depends strongly on the exponent distribution. Different exponent ratios produced different rates of improvement, and one sequence approached an energy-error plateau despite remaining below the chosen condition-number threshold. The plateau therefore could not be attributed to numerical instability on the available evidence alone. Increasing the number of functions does not necessarily improve the approximation efficiently if the added Gaussian widths do not address the remaining error.

The discrete parameter search demonstrated the benefit of selecting exponents rather than simply enlarging a fixed basis. Within the tested grid, five, six and eight Gaussians were the smallest bases satisfying relative energy-error targets of 10^{-2}, 10^{-3} and 10^{-4}, respectively. The selected eight-function basis achieved an error of 6.6406×10^{-5}, approximately 12.4 times smaller than that of the twelve-function reference basis, despite using four fewer functions. These results establish the effectiveness of exponent selection within the tested parameter ranges, rather than a globally optimal number of Gaussian functions. The selected exponent ratios also lay at the upper boundary of the search grid, indicating that the reported minimum basis sizes may depend on the chosen search limits.

The wavefunction calculations revealed a further distinction between energy accuracy and the accuracy of the state itself. For a normalised approximate ground state, the variational energy error depends quadratically on the excited-state components of the wavefunction error, with contributions weighted by their excitation energies. The energy can therefore converge more rapidly than the wavefunction-difference norm. At $N = 30$, the calculated L^2 error was approximately 1.505×10^{-5}, even though the energy error had decreased substantially. The near-nuclear calculations also confirmed a limitation inherent to finite expansions of centred spherical Gaussians. Each Gaussian has zero radial derivative at the nucleus, so no finite linear combination can reproduce the nonzero electron–nucleus cusp of the exact hydrogenic ground state. The resulting local energy diverges negatively near the nucleus, despite the total variational energy remaining finite. Improving the integrated energy therefore does not remove every local error in the wavefunction.

Numerical verification supported the consistency of the implementation through independent matrix-element integration, eigenpair-residual checks, independent L^2 integration and nuclear-charge scaling tests. These complementary checks demonstrated the importance of assessing numerical consistency alongside agreement with an exact benchmark. The conditioning calculations also demonstrated the importance of near-linear dependence as the basis grows. Canonical orthogonalisation can remove numerically sensitive directions, but doing so reduces the available variational space. Basis size, variational accuracy and numerical reliability must therefore be

considered together.

The nuclear-charge calculations provided a particularly clear example of why the physical length scale matters. Reusing the hydrogen exponents for the one-electron ion Mn^{24+} gave a relative energy error of approximately 54.9%, as the fixed basis failed to reproduce the strong contraction of the ground-state orbital. Scaling the Gaussian exponents by Z^2 reduced the error to the hydrogen value of approximately 0.0822% and recovered the contracted radial probability distribution without increasing the basis size. This behaviour follows directly from the coordinate scaling of the non-relativistic hydrogenic Hamiltonian and was confirmed numerically.

Beyond the individual numerical results, the investigation developed a deeper understanding of how physical accuracy, basis construction and numerical reliability interact in variational calculations. This project went beyond reproducing the hydrogen ground-state energy by examining wavefunction errors, the nuclear cusp, exponent selection and near-linear dependence. Implementing parameter searches, independent numerical checks and canonical orthogonalisation also provided practical experience in identifying and addressing limitations of scientific computations, rather than relying on a single measure of convergence.

The central conclusion is that there is no universal basis size that guarantees a given level of accuracy for every physical quantity or nuclear charge. A useful Gaussian basis must cover the relevant radial length scales, meet the accuracy requirements of the intended calculation and remain sufficiently well conditioned for a reliable variational solution. Physically appropriate exponent selection can therefore be more effective than increasing the number of functions alone.

5.2 Limitations and Further Work

The calculations were restricted to fixed-nucleus, non-relativistic, one-electron ground states represented by centred spherical Gaussian functions. This model permits direct comparison with exact hydrogenic solutions and isolates the effects of basis construction, but does not include electron–electron interactions, molecular bonding, relativistic effects or states requiring nonzero angular momentum. In particular, the exact Z^2 energy scaling and associated exponent scaling apply to the hydrogenic Hamiltonian used here. They cannot be assumed to hold unchanged for neutral atoms, many-electron ions or molecular systems, where additional interactions and length scales affect the wavefunctions.

A second limitation is the inability of a finite centred spherical Gaussian expansion to satisfy the exact electron–nucleus cusp condition. Although the basis can reproduce the overall energy and radial probability distribution closely, it retains an incorrect derivative at the nucleus and a divergent local energy. Future calculations could compare the Gaussian results with alternative radial functions that reproduce the cusp more naturally, or investigate explicitly cusp-corrected wavefunctions. Such comparisons would help determine how much the near-nuclear description can be improved without substantially increasing the computational cost.

The apparent plateau in the L^2 wavefunction error also remains unresolved. Independent integration methods agreed closely, supporting the numerical evaluation of the error, but this does not establish the cause of its convergence behaviour. A more systematic study could vary the diffuse and tight exponent limits independently, extend the basis-size range and compare different exponent distributions. Repeating selected calculations at higher numerical precision would also help distinguish basis limitations from floating-point sensitivity.

The reported minimum basis sizes are limited by the discrete parameter grid and the chosen condition-number screening criterion. Because several selected parameters occurred at the boundaries of the search range, extending the parameter domain would be particularly useful. Continuous optimisation of α and β could then be compared with the discrete search. It would also be valuable to optimise against different objectives, including the L^2 wavefunction error, mean-radius error or near-nuclear density, rather than the energy alone. This would test whether bases selected for minimum energy error are also suitable for other physical observables, or whether different accuracy requirements favour different exponent distributions.

The numerical conditioning analysis could also be extended. More systematic tests of canonical orthogonalisation thresholds, retained basis dimensions and higher-precision eigensolvers would help identify when near-linear dependence begins to limit the reliability of the calculated energy. Such tests would also clarify how much variational flexibility can be retained while excluding numerically unreliable directions.

Finally, the same approach could be extended to excited hydrogenic states, higher angular momenta and more general atomic or molecular Hamiltonians. These calculations would test how far the conclusions about exponent coverage and conditioning remain useful beyond the spherical one-electron ground state.

The hydrogenic model therefore provides a controlled demonstration of a broader principle in scientific computing: the quality of a numerical approximation depends on how well its mathematical representation captures the physical behaviour of the system and how reliably the solution is obtained. A successful variational basis must provide the required physical accuracy while remaining numerically stable, rather than being judged by the number of functions alone.

Appendix A

Derivation of Gaussian Matrix Elements

For the hydrogenic Hamiltonian in atomic units,

$$\hat{H} = -\frac{1}{2}\nabla^2 - \frac{Z}{r},$$

we use normalised spherical Gaussian functions,

$$\phi_i(r) = A_i e^{-\lambda_i r^2}, \qquad p_{ij} = \lambda_i + \lambda_j.$$

The matrix elements can be calculated analytically. Since the functions are spherically symmetric, the volume element becomes $d^3r = 4\pi r^2\, dr$ after integrating over the angles.

A.1 Gaussian Normalisation

Each Gaussian must satisfy

$$\int |\phi_i|^2\, d^3r = 1.$$

Substituting the Gaussian function gives

$$\begin{aligned} 1 &= 4\pi A_i^2 \int_0^\infty r^2 e^{-2\lambda_i r^2}\, dr \\ &= 4\pi A_i^2 \frac{\sqrt{\pi}}{4(2\lambda_i)^{3/2}} \\ &= A_i^2 \left(\frac{\pi}{2\lambda_i}\right)^{3/2}, \end{aligned}$$

using

$$\int_0^\infty r^2 e^{-ar^2}\, dr = \frac{\sqrt{\pi}}{4a^{3/2}}.$$

Solving for A_i gives

$$\boxed{A_i = \left(\frac{2\lambda_i}{\pi}\right)^{3/4}}. \tag{A.1}$$

A.2 Overlap Matrix

The overlap between two Gaussian functions is

$$\begin{aligned} S_{ij} &= \int \phi_i(\mathbf{r})\phi_j(\mathbf{r})\, d^3r \\ &= 4\pi A_i A_j \int_0^\infty r^2 e^{-p_{ij} r^2}\, dr \\ &= A_i A_j \frac{\pi^{3/2}}{p_{ij}^{3/2}}. \end{aligned}$$

Therefore,

$$\boxed{S_{ij} = A_i A_j \left(\frac{\pi}{p_{ij}}\right)^{3/2}}. \tag{A.2}$$

The diagonal elements are $S_{ii} = 1$ because each Gaussian is normalised. The off-diagonal elements are generally nonzero because the functions are not orthogonal.

A.3 Kinetic-Energy Matrix

The kinetic-energy matrix elements are

$$T_{ij} = -\frac{1}{2}\int \phi_i \nabla^2 \phi_j \, d^3r.$$

For a spherical Gaussian,

$$\begin{aligned}
\frac{d\phi_j}{dr} &= -2\lambda_j r \phi_j, \\
\nabla^2 \phi_j &= \frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{d\phi_j}{dr}\right) \\
&= (4\lambda_j^2 r^2 - 6\lambda_j)\phi_j.
\end{aligned}$$

Substituting this into the kinetic-energy integral,

$$T_{ij} = 3\lambda_j S_{ij} - 2\lambda_j^2 \int r^2 \phi_i \phi_j \, d^3r.$$

The remaining integral is

$$\begin{aligned}
\int r^2 \phi_i \phi_j \, d^3r &= 4\pi A_i A_j \int_0^\infty r^4 e^{-p_{ij} r^2} \, dr \\
&= 4\pi A_i A_j \frac{3\sqrt{\pi}}{8 p_{ij}^{5/2}} \\
&= \frac{3}{2p_{ij}} S_{ij}.
\end{aligned}$$

Substituting this result,

$$\begin{aligned}
T_{ij} &= 3\lambda_j S_{ij} - \frac{3\lambda_j^2}{p_{ij}} S_{ij} \\
&= \frac{3\lambda_j (p_{ij} - \lambda_j)}{p_{ij}} S_{ij} \\
&= \frac{3\lambda_i \lambda_j}{p_{ij}} S_{ij}.
\end{aligned}$$

Hence,

$$\boxed{T_{ij} = \frac{3\lambda_i \lambda_j}{p_{ij}} S_{ij}.} \tag{A.3}$$

This is symmetric in i and j, as expected for the kinetic-energy operator.

A.4 Electron–Nucleus Potential Matrix

The Coulomb potential is $V(r) = -Z/r$, giving

$$\begin{aligned}
V_{ij} &= -Z \int \frac{\phi_i \phi_j}{r} \, d^3r \\
&= -4\pi Z A_i A_j \int_0^\infty r e^{-p_{ij} r^2} \, dr \\
&= -4\pi Z A_i A_j \frac{1}{2p_{ij}},
\end{aligned}$$

using

$$\int_0^\infty r e^{-ar^2}\, dr = \frac{1}{2a}.$$

Therefore,

$$\boxed{V_{ij} = -\frac{2\pi Z A_i A_j}{p_{ij}}.} \tag{A.4}$$

A.5 Variational Eigenvalue Problem

The Hamiltonian matrix is obtained by adding the kinetic and potential contributions:

$$H_{ij} = \frac{3\lambda_i \lambda_j}{p_{ij}} S_{ij} - \frac{2\pi Z A_i A_j}{p_{ij}}. \tag{A.5}$$

Using the Gaussian expansion $\psi_{\mathrm{G}} = \sum_{i=1}^{N} c_i \phi_i$, the energy can be written as

$$E[\mathbf{c}] = \frac{\mathbf{c}^{\mathsf{T}} \mathbf{H} \mathbf{c}}{\mathbf{c}^{\mathsf{T}} \mathbf{S} \mathbf{c}}.$$

To find the coefficients that minimise the energy, we use the normalisation condition $\mathbf{c}^{\mathsf{T}} \mathbf{S} \mathbf{c} = 1$ and introduce a Lagrange multiplier E:

$$\mathcal{L}(\mathbf{c}, E) = \mathbf{c}^{\mathsf{T}} \mathbf{H} \mathbf{c} - E(\mathbf{c}^{\mathsf{T}} \mathbf{S} \mathbf{c} - 1).$$

Differentiating with respect to $\mathbf{c}$ and setting the result equal to zero,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{c}} &= 2\mathbf{H}\mathbf{c} - 2E\mathbf{S}\mathbf{c} = 0, \\ \mathbf{H}\mathbf{c} &= E\mathbf{S}\mathbf{c}. \end{aligned}$$

The lowest generalised eigenvalue gives the approximate ground-state energy within the chosen Gaussian basis. Since the matrix elements are calculated analytically, numerical integration is not needed to construct $\mathbf{H}$ or $\mathbf{S}$.

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